PROGRAMME FOR THE

JOINT VARENNA - LAUSANNE INTERNATIONAL WORKSHOP

"THEORY OF FUSION PLASMAS"

Villa Monastero, Varenna, Italy
21 - 25 September 2025

SUNDAY September 20th, 2026

16:00 - 19:00		Registration in Villa Monastero	

MONDAY, September 21st, 2026

9:00		*Welcome*	*G. Gorini and J. Graves*

Session 1: fundamental theory of turbulence and transport

9:10	I-1	A. Schekochihin	Tutorial: “As simple as possible but not simpler": physics of turbulence and transport via reduced fluid models
9:55	I-2	G. Acton	Predicting zonal flows in stellarators
10:40		*Coffee break*	
11:10	I-3	J. Burby	A statistical equilibrium model for stellarators
11:55	I-4	P. Costello	Constrained optimal modes: tightening thermodynamic bounds on the linear growth rate of microinstabilities
12:40		*Group Photo in Villa Monastero's garden*	
12:50		*Lunch*	

Session 2: tokamak plasma regimes

15:00	I-5	N. Rivals	Understanding of the X-Point Radiator regimes in tokamaks through an example in the WEST device
15:45	I-6	F. Maiden	The theory behind solenoid-free microwave driven start-up in tokamaks
16:30		*Coffee break*	
17:00	I-7	K. Zhang	Turbulence in the Quasicontinuous Exhaust Regime
17:45		*end*	
19:30		Welcome Party, Blanco Lounge Bar, Varenna	

TUESDAY, September 22nd, 2026

Session 3: fast ions, runaways and heat transport in stellarators

9:00	I-8	T. Foster	Alpha-particle orbits near rational flux surfaces in stellarators
9:45	I-9	M. Fitzgerald	Conservation and destruction of alpha particle invariants in optimized stellarators
10:30		*Coffee break*	
11:00	I-10	I. Ekmark	Runaway electrons in quasi-axisymmetric stellarators
11:45	I-11	H. Thienpondt	Thresholds for turbulent heat transport in Wendelstein 7-X: Insights from massive gyrokinetic simulations and experimental measurements
12:30		*Lunch*	
14:30		Poster session 1: P1 - P14	
16:00		*Coffee break*	
16:30		Poster session 2: P15 - P28	
18:00		*End poster sessions*	

WEDNESDAY, September 23rd, 2025

Session 4: microturbulent transport and gyrokinetics in tokamaks

9:00	I-12	H. Sun	The unusual dynamics of turbulent transport in spherical tokamaks with large field line pitch angle
9:45	I-13	L. De Gianni	Understanding Trapped Electron Modes in negative triangularity
10:30	I-14	S. Liu	Development of the Gyrokinetic-MHD Hybrid Code cuGMEC and Its Nonlinear Simulations of Alpha Particle-driven Alfven Eigenmodes in ITER
11:15		*Coffee break*	
11:35	I-15	S. Choudhary	Gyrokinetic Transport in flat-Temperature Tokamak Plasmas
12:20	I-16	J. McClenaghan	Pedestal Stability and KBM/MTM Transport Limits in Tokamaks using QLGYRO
13:05		*Lunch*	
19:30		Conference dinner: Ristorante La Breva in Lierna (dinner at 20.00, bus at 19.30)	

THURSDAY, September 24th, 2025

Session 5: stellarator optimisation

9:00	I-17	J. Velasco,	Tutorial: New avenues for stellarator design enabled by piecewise omnigenity
9:45	I-18	W. Sengupta	Periodic Korteweg-de Vries soliton potentials generate quasisymmetric magnetic field strength in both vacuum and finite plasma-beta equilibrium
10:30		*Coffee break*	
11:00	I-19	A. Goodman	Searching for SQuIDs: Developments in the optimisation of Stable Quasi-Isodynamic Designs
11:45	I-20	P. Jurašić	Designing stellarator island divertors by controlling the island flux
12:30		*Lunch*	
14:30			Poster session 3: P29 - P42
16:00		*Coffee break*	
16:30			Poster session 4: P43 - P56
18:00		*End poster sessions*	

FRIDAY, September 25th, 2026

Session 6: fast ion and nonlinear physics in tokamaks

9:00	I-21	H. Cai	Multiscale gyrokinetic simulations of the interaction between turbulence and fishbones
9:45	I-22	D. Korger	A nonlinear Schrodinger equation for the description of geodesic-acoustic-modes in tokamaks
10:30		*Coffee break*	
11:00	I-23	J. Wang	Comprehensive simulations of bursting shear Alfven waves in tokamaks and their control via phase space engineering with ICRF waves
11:45	I-24	E. G. Devin	Emergence of persistent advection-diffusion transport structure and nonlinear amplitude evolution of strongly driven instabilities
12:30		*Closing session*	
12:45		*end*	